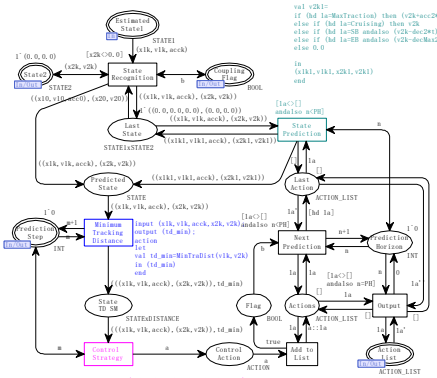
1. CPN model of the Coupling Scenario




```
input (x1k,v1k,acck,x2k,v2k,la);
output (x1k1,v1k1,x2k1,v2k1);
action
let
val x1k1=
if (acck<0.0) then (x1k+v1k*t+0.5*acck*t*t)
else if (acck>0.0-0.00001 andalso acck<0.0+0.00001) then (x1k+v1k*t)
else if (acck<0.0 andalso (v1k+acck*t)>0.0) then (x1k+v1k*t+0.5*acck*t*t)
else x1k+0.5*v1k*v1k/acck

val v1k1=
if (acck<0.0) then (v1k+acck*t)
else if (acck>0.0-0.00001 andalso acck<0.0+0.00001) then v1k
else if (acck<0.0 andalso (v1k+acck*t)>0.0) then (v1k+acck*t)
else 0.0

val x2k1=
if (hd la=MaxTraction) then (x2k+v2k*t+0.5*acc2*t*t)
else if (hd la=Cruising) then (x2k+v2k*t)
else if (hd la=SB andalso (v2k-dec2*t)>0.0) then (x2k+v2k*t-0.5*dec2*t*t)
else if (hd la=SB andalso (v2k-dec2*t)<0.0)) then (x2k+0.5*v2k*v2k/dec2)
else if (hd la=EB andalso (v2k-decMax2*t)>0.0) then (x2k+v2k*t-0.5*decMax2*t*t)
else x2k+0.5*v2k*v2k/decMax2

val v2k1=
if (hd la=MaxTraction) then (v2k+acc2*t)
else if (hd la=Cruising) then v2k
else if (hd la=SB andalso (v2k-dec2*t)>0.0) then (v2k-dec2*t)
else if (hd la=EB andalso (v2k-decMax2*t)>0.0) then (v2k-decMax2*t)
else 0.0

in
(x1k1,v1k1,x2k1,v2k1)
end
```

```
input (x1k,v1k,acck,x2k,v2k,td_min,m);
output (a);
action
let
val real_m =  Real.fromInt m
val a =

if (((x1k-x2k)>=td_min)
andalso v1k>0.0 andalso v1k+acck*t>0.0
andalso (((x1k+v1k*t+0.5*acck*t*t)-(x2k+v2k*t+0.5*acc2*t*t))>=MinTraDist(v1k+acck*t, v2k+acc2*t))
andalso ((v2k+acc2*t)<=v_limit2))
then MaxTraction

else if (((x1k-x2k)>=td_min)
andalso v1k>0.0 andalso v1k+acck*t>0.0
andalso (((x1k+v1k*t+0.5*acck*t*t)-(x2k+v2k*t+0.5*acc2*t*t))>=MinTraDist(v1k+acck*t, v2k+acc2*t))
andalso ((v2k+acc2*t)>v_limit2)
andalso (v2k<=v_limit2))
then Cruising

else if (((x1k-x2k)>=td_min)
andalso v1k>0.0 andalso v1k+acck*t>0.0
andalso (((x1k+v1k*t+0.5*acck*t*t)-(x2k+v2k*t+0.5*acc2*t*t))<MinTraDist(v1k+acck*t, v2k+acc2*t))
andalso (((x1k+v1k*t+0.5*acck*t*t)-(x2k+v2k*t))>=MinTraDist(v1k+acck*t, v2k)))
then Cruising

else if (((x1k-x2k)>=td_min)
andalso v1k>0.0 andalso v1k+acck*t>0.0
andalso (((x1k+v1k*t+0.5*acck*t*t)-(x2k+v2k*t))<MinTraDist(v1k+acck*t, v2k)))
andalso v2k-dec2*t>0.0
andalso (((x1k+v1k*t+0.5*acck*t*t)-(x2k+v2k*t-0.5*dec2*t*t))>=MinTraDist(v1k+acck*t, v2k-dec2*t)))
then SB

else if (((x1k-x2k)>=td_min)
andalso v1k>0.0 andalso v1k+acck*t>0.0
andalso (((x1k+v1k*t+0.5*acck*t*t)-(x2k+v2k*t))<MinTraDist(v1k+acck*t, v2k)))
andalso v2k-dec2*t>0.0
andalso (((x1k+v1k*t+0.5*acck*t*t)-(x2k+0.5*v2k*v2k/dec2))>=MinTraDist(v1k+acck*t, 0.0)))
then SB

else if (((x1k-x2k)>=td_min)
andalso v1k>0.0 andalso v1k+acck*t>0.0
andalso v2k-dec2*t>0.0
andalso (((x1k+v1k*t+0.5*acck*t*t)-(x2k+v2k*t-0.5*dec2*t*t))<MinTraDist(v1k+acck*t, v2k-dec2*t)))
then EB

else if (((x1k-x2k)>=td_min)
andalso v1k>0.0 andalso v1k+acck*t>0.0
andalso v2k-dec2*t<0.0
andalso (((x1k+v1k*t+0.5*acck*t*t)-(x2k+0.5*v2k*v2k/dec2))<MinTraDist(v1k+acck*t, 0.0)))
then EB

else if (((x1k-x2k)>=td_min)
andalso v1k>0.0 andalso v1k+acck*t<0.0
andalso (((x1k+0.5*v1k*v1k/acck)-(x2k+v2k*t+0.5*acc2*t*t))>=MinTraDist(0.0, v2k+acc2*t))
andalso ((v2k+acc2*t)<=v_limit2))
then MaxTraction

else if (((x1k-x2k)>=td_min)
andalso v1k>0.0 andalso v1k+acck*t<0.0
andalso (((x1k+0.5*v1k*v1k/acck)-(x2k+v2k*t+0.5*acc2*t*t))>=MinTraDist(0.0, v2k+acc2*t))
andalso ((v2k+acc2*t)>v_limit2)
andalso (v2k<=v_limit2))
then Cruising

else if (((x1k-x2k)>=td_min)
andalso v1k>0.0 andalso v1k+acck*t<0.0
andalso (((x1k+0.5*v1k*v1k/acck)-(x2k+v2k*t+0.5*acc2*t*t))>=MinTraDist(0.0, v2k+acc2*t))
andalso (((x1k+0.5*v1k*v1k/acck)-(x2k+v2k*t))<MinTraDist(0.0, v2k)))
then Cruising

else if (((x1k-x2k)>=td_min)
andalso v1k>0.0 andalso v1k+acck*t<0.0
andalso (((x1k+0.5*v1k*v1k/acck)-(x2k+v2k*t))<MinTraDist(0.0, v2k))
andalso v2k-dec2*t>0.0
andalso (((x1k+0.5*v1k*v1k/acck)-(x2k+v2k*t-0.5*dec2*t*t))>=MinTraDist(0.0, v2k-dec2*t)))
then SB

else if (((x1k-x2k)>=td_min)
andalso v1k>0.0 andalso v1k+acck*t<0.0
andalso (((x1k+0.5*v1k*v1k/acck)-(x2k+v2k*t))<MinTraDist(0.0, v2k))
andalso v2k-dec2*t<0.0
andalso (((x1k+0.5*v1k*v1k/acck)-(x2k+0.5*v2k*v2k/dec2))>=MinTraDist(0.0, 0.0)))
then SB

else if (((x1k-x2k)>=td_min)
andalso v1k>0.0 andalso v1k+acck*t<0.0
andalso v2k-dec2*t<0.0
andalso (((x1k+0.5*v1k*v1k/acck)-(x2k+v2k*t-0.5*dec2*t*t))<MinTraDist(0.0, v2k-dec2*t)))
then EB

else if (((x1k-x2k)>=td_min)
andalso v1k>0.0 andalso v1k+acck*t<0.0
andalso v2k-dec2*t<0.0
andalso (((x1k+0.5*v1k*v1k/acck)-(x2k+0.5*v2k*v2k/dec2))<MinTraDist(0.0, 0.0)))
then EB

else if (((x1k-x2k)>=td_min)
andalso v1k>0.0-0.00001 andalso v1k<0.0+0.00001
andalso acck<0.0
andalso  (((x1k-(x2k+v2k*t+0.5*acc2*t*t))>=MinTraDist(v1k, v2k+acc2*t))
andalso ((v2k+acc2*t)<=v_limit2))
then MaxTraction

else if (((x1k-x2k)>=td_min)
andalso v1k>0.0-0.00001 andalso v1k<0.0+0.00001
andalso acck>0.0
andalso  (((x1k-(x2k+v2k*t+0.5*acc2*t*t))>=MinTraDist(v1k, v2k+acc2*t))
andalso ((v2k+acc2*t)>v_limit2)
andalso (v2k<=v_limit2))
then Cruising

else if (((x1k-x2k)>=td_min)
andalso v1k>0.0-0.00001 andalso v1k<0.0+0.00001
andalso acck<0.0
andalso  (((x1k-(x2k+v2k*t+0.5*acc2*t*t))<MinTraDist(v1k, v2k+acc2*t))
andalso (((x1k-(x2k+v2k*t))>=MinTraDist(v1k, v2k+0.1))
then Cruising

else if (((x1k-x2k)>=td_min)
andalso v1k>0.0-0.00001 andalso v1k<0.0+0.00001
andalso acck<0.0
andalso (((x1k-(x2k+v2k*t))<MinTraDist(v1k, v2k)+0.1)
andalso v2k-dec2*t<0.0
andalso (((x1k-(x2k+v2k*t-0.5*dec2*t*t)))>=MinTraDist(v1k, v2k-dec2*t)))
then SB

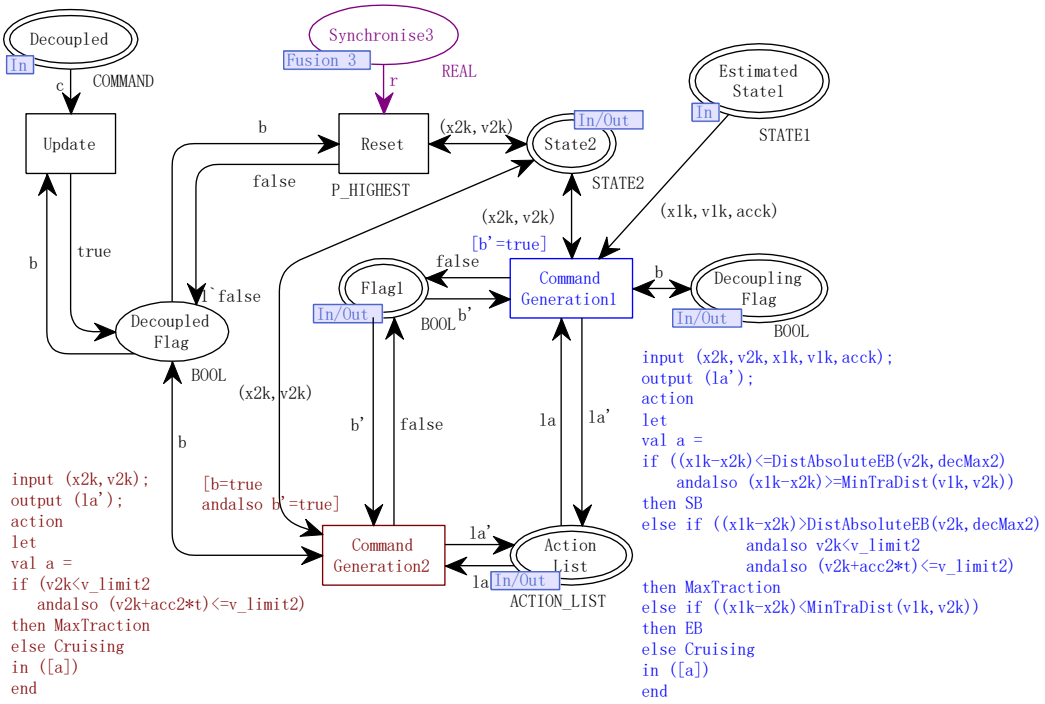
else if (((x1k-x2k)>=td_min)
andalso v1k>0.0-0.00001 andalso v1k<0.0+0.00001
andalso acck<0.0
andalso (((x1k-(x2k+v2k*t))<MinTraDist(v1k, v2k)+0.1)
andalso v2k-dec2*t<0.0
andalso (((x1k-(x2k+0.5*v2k*v2k/dec2)))>=MinTraDist(v1k, 0.0)))
then SB

else if (((x1k-x2k)>=td_min)
andalso v1k>0.0-0.00001 andalso v1k<0.0+0.00001
andalso acck<0.0
andalso v2k-dec2*t>0.0
andalso (((x1k-(x2k+v2k*t-0.5*dec2*t*t))<MinTraDist(v1k, v2k-dec2*t)))
then EB

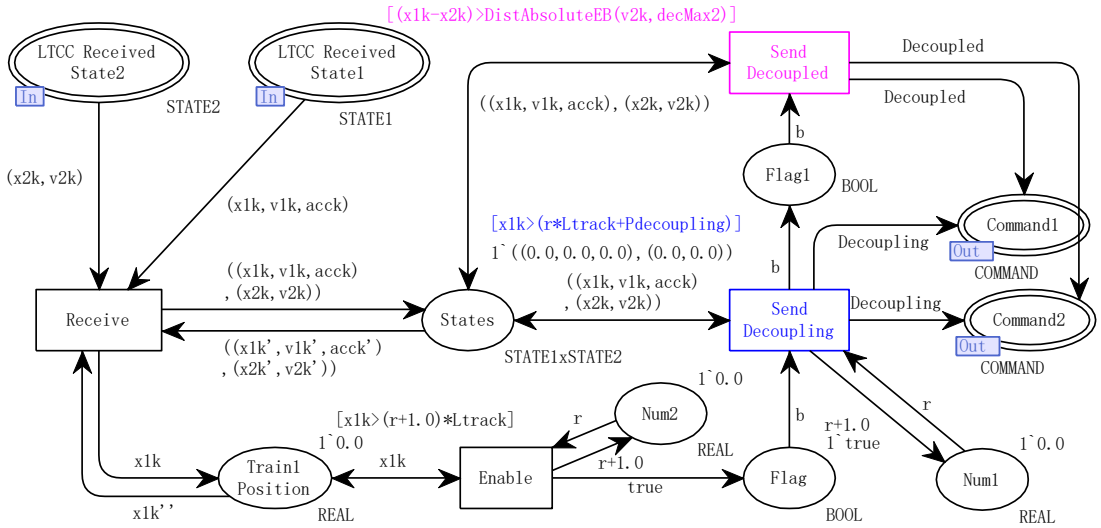
else if (((x1k-x2k)>=td_min)
andalso v1k>0.0-0.00001 andalso v1k<0.0+0.00001
andalso acck<0.0
andalso v2k-dec2*t<0.0
andalso (((x1k-(x2k+0.5*v2k*v2k/dec2))<MinTraDist(v1k, 0.0)))
then EB

else EB
in (a)
end
```

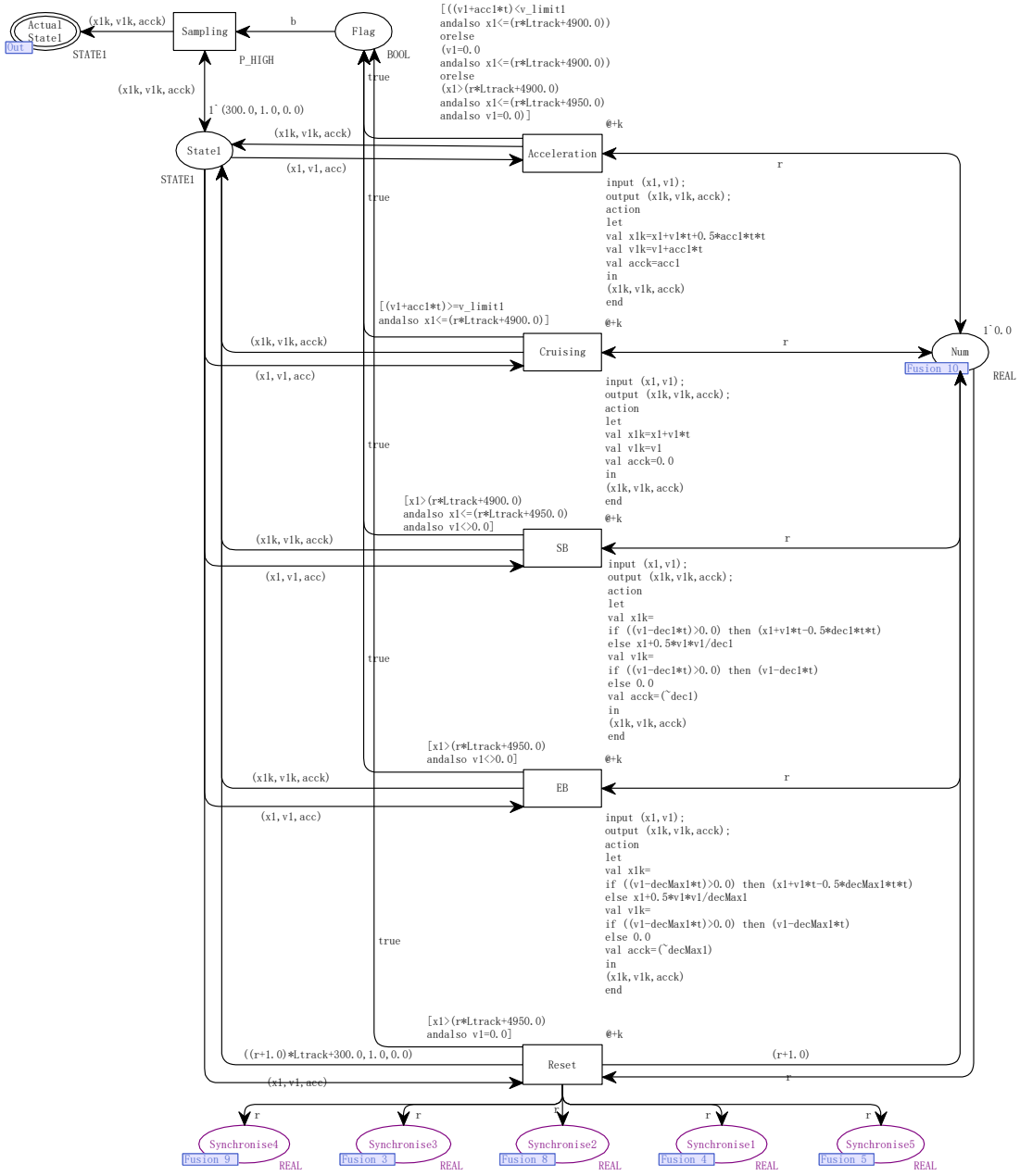
2. CPN model of the Decoupling Scenario



3. CPN model of the LTCC



4. CPN model of the dynamics of the preceding train



5. CPN model of the On-board controller of the preceding train

